F78PB Probability and Statistics B

Tutorial 1: Moment generating functions

1. Suppose X is a random variable with density given by

$$f(x) = 2e^{-2x+2} \quad \text{for} \quad x \geq 1$$

(and $f(x) = 0$ otherwise). Determine $M_X(\theta)$, the moment generating function for X, and the values of θ for which $M_X(\theta)$ is defined. Use $M_X(\theta)$ to determine $\mathbb{E}(X^2)$.

2. Suppose that X is a random variable with density function given by

$$f_X(x) = \begin{cases} \frac{1}{2}e^{x} & x < 0, \\ \frac{1}{2}e^{-x} & x \geq 0. \end{cases}$$

(a) Show that the moment generating function, $M_X(\theta)$, for X is given by

$$M_X(\theta) = \frac{1}{1-\theta^2} \qquad \text{for } -1 < \theta < 1.$$

(b) Use the series expansion for $M_X(\theta)$ to determine $\mathbb{E}(X^4)$.

3. Suppose that $X \sim \text{Gamma}\,(t, \lambda)$ where $t, \lambda > 0$.

(a) Show that the moment generating function $M_X(\theta)$ given by

$$M_X(\theta) = \left(\frac{\lambda}{\lambda - \theta}\right)^t$$

for $\theta < \lambda$.

(b) Suppose that $X \sim \text{Gamma}\,(2, \lambda)$. Use the power series expansion for $M_X(\theta)$ to determine $\mathbb{E}(X^4)$.

(c) Suppose that $X_1, X_2, ..., X_n$ are *independent* and *identically distributed* random variables, each with a $\text{Gamma}\,(t, \lambda)$ distribution, and let $Z = X_1 + X_2 + ... + X_n$. Determine $M_Z(\theta)$, the moment generating function of Z and hence, or otherwise, determine the distribution of Z.

4. Suppose that $X \sim \text{Gamma}\,(t, \lambda)$ where $t, \lambda > 0$ and let $Y = cX$ where c is a positive constant. Compute the moment generating function, $M_Y(\theta)$, of Y and hence or otherwise, determine the distribution of Y.

5. Let X be normally distributed with mean 3 and variance 1. Use $M_X(\theta)$, the moment generating function of X, to find $\mathbb{E}(X^3)$ and $\mathbb{E}(X^4)$.

Hint: Recall that $M_X(\theta) = e^{3\theta+\theta^2/2} = e^{\mu\theta}e^{\theta^2/2}$ and that the power series expansion for $M_X(\theta)$ is given by $M_X(\theta) = 1 + \mathbb{E}(X)\theta + \frac{\mathbb{E}(X^2)\theta^2}{2!} + \frac{\mathbb{E}(X^3)\theta^3}{3!} + ...$.

6. Suppose that $X \sim \text{N}\left(\mu_X, \sigma_X^2\right)$ and $Y \sim \text{N}\left(\mu_Y, \sigma_Y^2\right)$ are *independent* random variables. Let $S = aX + bY$ for some $a, b \in \mathbb{R}$, and use a moment generating argument to show that $S \sim \text{N}\left(a\mu_X + b\mu_Y, a^2\sigma_X^2 + b^2\sigma_Y^2\right)$.
Remark This shows that a linear combination of independent normal random variables is also a normal random variable.

Application
Anne and David agree to meet at a restaurant for lunch. Suppose that X, the time that Anne arrives at the restaurant in minutes past noon, is normally distributed with mean 0 and standard deviation 3 minutes and suppose that Y, the time that David arrives at the restaurant in minutes past noon, is normally distributed with mean 1 and standard deviation 4 minutes. Determine the probability that Anne must wait more than 4 minutes at the restaurant for David to arrive.

7. Suppose that $X_1, X_2, ...$ are i.i.d. random variables with common moment generating function $M_X(\theta)$ and suppose that N is a counting variable with probability generating function $G_N(s)$. In addition, suppose that N is independent of $X_1, X_2, ...$ and let $S_N = X_1 + X_2 + ... + X_N$. Use the moment generating function for S_N to show that $\mathbb{E}(S_N) = \mathbb{E}(N)\mathbb{E}(X)$.

8. A radio station is running a phone-in contest. Suppose that the probability that any given caller answers the question correctly is 1/5 (independently of all other callers) and suppose that the times between consecutive calls (in minutes) to the station are i.i.d. exponential random variables with parameter $\lambda = 2$. The DJ at the radio station starts to answer the phone at 9AM. Determine the distribution of the length of time, T, that elapses until the first correct answer is received by the station. (You may assume that the time until the DJ answers the first call has an Exp(2) distribution and is independent of the times between all subsequent calls to the station.)

9. A random variable X takes on the values $-1, 0,$ and 1 with probabilities $w, 1 - 2w,$ and w respectively. Let $Y = aX$ for $\{a \in \mathbb{R}, a \neq 0\}$.

   (a) Find all possible values of w such that Y admits a legitimate probability mass function.

   (b) What is the cumulant generating function of Y?

   (c) Find the mean and variance of Y using the cumulant generating function obtained from (b).

   (d) The standard measure of skewness is based on the second and third cumulants

   $$\gamma_1 = \frac{\kappa_3}{(\kappa_2)^{\frac{3}{2}}}.$$

   What is the measure of skewness γ_1 for Y?

10. Consider two independent random variables X_1 and X_2. Show that the cumulant generating function (cgf) of $Y = X_1 + X_2$ is

$$K_Y(\theta) = K_{X_1}(\theta) + K_{X_2}(\theta).$$

Exam Style Question

Claim sizes in a certain insurance situation are modelled by a distribution with moment generating function $M_X(\theta)$ given by

$$M_X(\theta) = (1 - 10\theta)^{-2}.$$

(a) Use the equation that links expected values to the derivatives of moment generating functions to show that $\mathbb{E}(X^2) = 600$. Also find the value of $\mathbb{E}(X^3)$. (4 marks)

(b) Suppose that W is a U(0, 1) random variable and suppose that $Y = -\log(W)$. Let $M_Y(\theta)$ be the moment generating function of Y.

   (i) Show that $M_Y(\theta) = \frac{1}{1-\theta}$ for $\theta < 1$ and determine the distribution of Y. (4 marks)

   (ii) Determine $\mathbb{E}[(\log(W))^2]$. (2 marks)

[Total 10 marks]